Table 7. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), multi-vehicle crashes without a motorcycle, Thailand, 2016–2023 (yearly periods), adaptive fused temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016]	–0.247	–1.76		
Constant [2017]	–0.360	–2.72		
Constant [2018–2020]	–0.629	–5.79		
Constant [2021]	–0.426	–3.36		
Constant [2022–2023]	–0.521	–4.31		
Variables producing the same parameter value across all periods [2016–2023]				
Vehicle involvement				
Two or more pickup trucks involved (1 if yes, 0 otherwise) [2016–2023]	–1.813	–11.20	0.0976	–0.0976
(Standard deviation of parameter distribution; 13.5% above zero)	1.647	8.93		
One 6-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.431	–3.34	0.0046	–0.0046
(Standard deviation of parameter distribution; 38.5% above zero)	1.472	6.03		
One 10-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.424	–2.94	–0.0072	0.0072
(Standard deviation of parameter distribution; 39.7% above zero)	1.619	5.90		
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	–0.309	–2.82	–0.0243	0.0243
(Standard deviation of parameter distribution; 42.6% above zero)	1.656	9.14		
Two or more trucks with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	–1.841	–3.66	0.0286	–0.0286
(Standard deviation of parameter distribution; 27.4% above zero)	3.062	4.50		
Bus involved (1 if yes, 0 otherwise) [2016–2023]	–0.210	–1.75	0.0174	–0.0174
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2016–2023]	–0.436	–8.23	0.0376	–0.0376
Roadway characteristics				
Concrete pavement surface (1 if yes, 0 otherwise) [2016–2023]	–0.178	–2.74	0.0149	–0.0149
On grade (1 if yes, 0 otherwise) [2016–2023]	–0.346	–2.16	0.0279	–0.0279
Temporal characteristics				
Midnight / early morning (1 if yes, 0 otherwise) [2016–2023]	0.448	7.05	–0.0407	0.0407
Peak hours (1 if yes, 0 otherwise) [2016–2023]	0.116	1.93	–0.0101	0.0101
Region				
Eastern region (1 if yes, 0 otherwise) [2016–2023]	0.570	8.38	–0.0526	0.0526
Western region (1 if yes, 0 otherwise) [2016–2023]	0.585	6.97	–0.0550	0.0550
Variables with temporally varying parameters				
Vehicle involvement				
One passenger car involved (1 if yes, 0 otherwise) [2016–2017]	–0.966	–9.07	0.0943	–0.0943
One passenger car involved (1 if yes, 0 otherwise) [2018–2023]	–0.728	–11.20	0.0589	–0.0589
Two or more passenger cars involved (1 if yes, 0 otherwise) [2016–2017]	–1.894	–13.27	0.1683	–0.1683
Two or more passenger cars involved (1 if yes, 0 otherwise) [2018–2019]	–1.182	–9.44	0.1026	–0.1026
Two or more passenger cars involved (1 if yes, 0 otherwise) [2020–2023]	–1.454	–13.28	0.0947	–0.0947
One pickup truck involved (1 if yes, 0 otherwise) [2016–2021]	–0.570	–8.77	0.0511	–0.0511
One pickup truck involved (1 if yes, 0 otherwise) [2022–2023]	–0.411	–4.42	0.0305	–0.0305
Crash type				
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2016–2019]	–0.830	–7.37	0.0723	–0.0723
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2020–2023]	–2.220	–4.23	0.0383	–0.0383
(Standard deviation of parameter distribution; 23.7% above zero)	3.098	5.43		
Head-on collision (1 if yes, 0 otherwise) [2016–2020]	1.264	12.78	–0.1437	0.1437
Head-on collision (1 if yes, 0 otherwise) [2021–2023]	1.027	9.70	–0.0916	0.0916
Crash cause				
Cutting in too closely (1 if yes, 0 otherwise) [2016–2021]	–0.986	–4.11	0.0225	–0.0225
(Standard deviation of parameter distribution; 31.3% above zero)	2.030	5.87		
Driver fell asleep (1 if yes, 0 otherwise) [2016–2020]	0.681	5.52	–0.0709	0.0709

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Driver fell asleep (1 if yes, 0 otherwise) [2021–2023]	0.502	3.06	–0.0415	0.0415
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2017]	–0.556	–3.57	0.0516	–0.0516
Road under construction (1 if yes, 0 otherwise) [2016–2019]	0.749	3.97	–0.0828	0.0828
Painted median (1 if yes, 0 otherwise) [2016–2020]	0.358	3.56	–0.0354	0.0354
Raised or earth median (1 if yes, 0 otherwise) [2016–2018]	–0.498	–5.39	0.0482	–0.0482
Raised or earth median (1 if yes, 0 otherwise) [2021–2022]	–0.320	–3.12	0.0242	–0.0242
Raised or earth median (1 if yes, 0 otherwise) [2023]	–0.597	–4.41	0.0407	–0.0407
Depressed median (1 if yes, 0 otherwise) [2016–2018]	–0.436	–5.07	0.0425	–0.0425
Depressed median (1 if yes, 0 otherwise) [2019–2023]	–0.210	–2.97	0.0163	–0.0163
Barrier median (1 if yes, 0 otherwise) [2016–2018]	–1.160	–9.82	0.1040	–0.1040
Barrier median (1 if yes, 0 otherwise) [2019–2023]	–2.036	–20.82	0.1422	–0.1422
Horizontal curve (1 if yes, 0 otherwise) [2016]	0.578	3.08	–0.0634	0.0634
Horizontal curve (1 if yes, 0 otherwise) [2017–2023]	–0.545	–2.91	–0.0135	0.0135
(Standard deviation of parameter distribution; 40.3% above zero)	2.229	6.19		
T-intersection (1 if yes, 0 otherwise) [2018–2023]	–0.339	–3.04	0.0264	–0.0264
Four-leg intersection (1 if yes, 0 otherwise) [2016]	–0.513	–2.12	0.0477	–0.0477
Four-leg intersection (1 if yes, 0 otherwise) [2017–2018]	0.375	2.19	–0.0399	0.0399
Four-leg intersection (1 if yes, 0 otherwise) [2019–2023]	–0.317	–2.84	0.0239	–0.0239
Environmental characteristics				
Wet road surface (1 if yes, 0 otherwise) [2016]	–0.422	–1.97	0.0398	–0.0398
Wet road surface (1 if yes, 0 otherwise) [2018–2020]	–0.450	–3.95	0.0379	–0.0379
Wet road surface (1 if yes, 0 otherwise) [2021–2023]	–0.557	–4.34	0.0385	–0.0385
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2019]	1.289	12.32	–0.1533	0.1533
Dark, no street lighting (1 if yes, 0 otherwise) [2020–2023]	0.773	7.08	–0.0675	0.0675
Dark, with street lighting (1 if yes, 0 otherwise) [2016–2020]	0.499	7.42	–0.0480	0.0480
Dark, with street lighting (1 if yes, 0 otherwise) [2021–2023]	0.319	3.82	–0.0247	0.0247
Temporal characteristics				
New Year festival period (1 if yes, 0 otherwise) [2019–2023]	–0.369	–3.50	0.0275	–0.0275
Region				
Northeastern region (1 if yes, 0 otherwise) [2016]	0.870	5.99	–0.0974	0.0974
Northeastern region (1 if yes, 0 otherwise) [2017–2023]	0.659	10.66	–0.0604	0.0604
Northern region (1 if yes, 0 otherwise) [2021–2023]	–0.491	–3.47	0.0342	–0.0342
Southern region (1 if yes, 0 otherwise) [2016–2020]	0.507	5.99	–0.0506	0.0506
Heterogeneity in the means of random parameters				
One truck with more than 10 wheels involved [2016–2023]: Evening [2017–2023]	0.572	4.34		
Cutting in too closely [2016–2021]: T-intersection [2016–2017]	1.490	3.36		
Horizontal curve [2017–2023]: Raining [2020–2023]	–1.144	–3.24		
Heterogeneity in the variances of random parameters				
One 10-wheel truck involved [2016–2023]: Songkran festival period [2016–2021]	1.315	2.37		
Model statistics				
Number of observations		33,646		
Number of estimated parameters		72		
Log-likelihood at zero, LL(0)		–23321.63		
Log-likelihood with constant only, LL(C)		–15177.81		
Log-likelihood at convergence, LL(β)		–12617.43		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.459		
AIC / BIC		25,378.9 / 25,985.4		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure re-selected by the adaptive fused lasso inside the random parameters model: the sparsest penalty whose structure is not rejected at the 90% level against the fully unconstrained (separate period) random parameters models ($LR < \chi^2_{0.90}(df)$); candidate structures were rebuilt and pruned in order of sparsity until one was not rejected (Table D1). The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully

unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = –(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.